MIL-E-1/1514A (NAVY)
8 February 1966

SUPERSEDING
MIL-E-1/1514 (NAVY)
18 December 1963
(See Note 14)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CLIPPER, HYDROGEN

TYPE 8492

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Clipper, Hydrogen

DIMENSIONS AND PIN CONNECTIONS: (See figure 1).

MOUNTING POSITION: Horizontal, or vertical with base down.

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit	Ef Vac	epx kv	ib a	Ib Adc	Ip Aac	Fault Current (0.15 sec max) a
Line Clipper:	5.0 ± 5%	33	500	1	20	1500
TEST CONDITIONS:	5.0	---	---	---	---	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:	tk sec	Eres Vac	Alt ft	TA °C	TE °C (Note 1)
Line Clipper:	180	5.0 ± 5%	10,000	+125-50	400
TEST CONDITIONS:	---	5.0	---	---	---

REQUIREMENT OR TEST:

GENERAL

3.1.1 Performance (Note 2)
3.2 Qualification - Required (Note 12)
Section 5. Preparation for delivery (Note 13)
E-50.2 Holding period - t = 96 hrs.
E-50.3 Preheating

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	AQL (Percent Defective)	INSP. LEVEL OR CODE	SYMBOL	LIMITS		UNITS
						MIN	MAX	
	Quality conformance inspection, part 1							
Appendix D, 20	Visual and mechanical inspection criteria							
1231	Emission	Ib = 8 Adc; f = 60 cps	0.65	II	etd	---	70	v
1301	Cathode heater current		0.65	II	If	12	15	Aac
1301	Reservoir heater current		0.65	II	Ires	2	3	Aac
---	Operation (1)	(Notes 5 and 6) Ef = Eres = 5.00 Vac	0.65	II	E_z	---	500	v
---	Operation (2)	(Notes 4 and 5) Ef = Eres = 5.25 Vac	0.65	II	---	---	---	---
---	Operation (3)	(Notes 4 and 5) Ef = Eres - 4.75 Vac	0.65	II	---	---	---	---
	Quality conformance inspection, part 2							
4.1.1.2	Periodic check tests							
1031	Sweep frequency vibration	f = 50 to 1000 to 50 cps; g = 10; planes X, Y, and Z; t = 10 minutes per plane, 30 minutes total		Note 3				
---	Post sweep frequency vibration end point	Operation (1)						
1031	Variable frequency vibration	No voltages; (Note 3) f = 5 to 50 to 5 cps	6.5	L6				
---	Long duration shock	Shock = 11 ± 2 ms; acceleration = 50 g peak min; 3 planes; Ef = Eres = 5.0 vac; (Note 7)		Notes 8 and 9				
---	Post variable frequency vibration and shock end point	Operation (1)						

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	AQL (Percent Defective)	INSP. LEVEL OR CODE	SYMBOL	LIMITS		UNITS
						MIN	MAX	
	Quality conformance inspection, part 3							
4.7	Life test	Group C; (Notes 10 and 11); Operation (1); R_1 adjusted for ib = 500 a; prr = 310 pps			t	500	---	hrs
4.7.3	Life test end point	Operation (1)			E_Z	---	750	v

Note 1: No external portion of the tube shall be permitted to exceed this maximum temperature.

Note 2: In addition to the paragraphs specified herein, the following tests and requirements listed in 3.1.1 shall apply: 3.5, 3.6.1, 3.7, 3.8, 4.1, E-40 and E-50.

Note 3: This test shall be performed on a minimum of two tubes.

Note 4: The tube shall go into conduction smoothly and operate for 30 minutes without arcback. Three arcbacks are permitted prior to an uninterrupted 30 minute operation run.

Note 5: The tube shall be operated as a line clipper in a thyratron modulator (see figure 2) with the following circuit constants: peak thyratron forward voltage = 33 kv, R_L adjusted to give 5 percent negative mismatch, t_p = 2.5 ± 0.25 us, prr = 500 pps, R_2 = 8.5 ohms, Z_o = 8.5 ohms.

Note 6: The tube shall be operated as a line clipper in a thyratron modulator (see figure 2). Raise voltage applied to clipper to value at which tube goes into conduction and record this value. This shall be the first test after the holding period. Then tube shall be operated 30 minutes under conditions specified in note 5.

Note 7: The tubes shall be subjected to one long duration shock in the direction of each of the three mutually perpendicular axis (X, Y, and Z) for a total of six blows. Electrical discontinuities shall be monitored. This is a non-destructive test.

The following test shall be performed in each 18-month interval: The tube shall be subjected to seven long duration shocks in the direction of each of the three mutually perpendicular axis (X, Y, and Z) for a total of 21 blows. Electrical discontinuities shall be monitored. This is a destructive test.

Note 8: Shock test sampling and acceptance criteria shall be as follows: For one-month's production to be sampled up to 250 units, the first sample (n1) shall consist of one tube with an acceptance number (c1) of zero (0). Second sample (n2) shall consist of two tubes with an acceptance number (c2) equal to (1) for the combined sample.

Note 9: This test shall be performed on a minimum of three tubes for qualification.

Note 10: When one-month's production to be sampled is 50 units or less, paragraph 4.7.4.4 shall apply. When one-month's production to be sampled is more than 50 units but less than 250 units, the following alternate sampling procedure may be used:

n1 = 2 c1 = 0
n2 = 2 c2 = 1

except that c1 = 1 if the first sample from the preceding lot contained no defectives.

Note 11: Life test shall be shut down every 96 hours for a 60-minute interval.

Note 12: The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be

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obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

Note 13: Preservation, packaging and packing. - Unless otherwise specified in the contract or order, preservation, packaging and packing shall be as follows:

(a) Preservation and packaging shall be sufficient to afford adequate protection against corrosion and deterioration during shipment from the supply source to the using activity and until installation.

(b) Packing shall be accomplished in a manner which will insure acceptance and protection against physical or mechanical damage during direct shipment from the supply source to the using activity.

Note 14: CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Review activity:
Navy - SH

User activities:
Navy, WP, MC, CG

Preparing activity:
Navy - SH
(Project 5960-N645 (NAVY))

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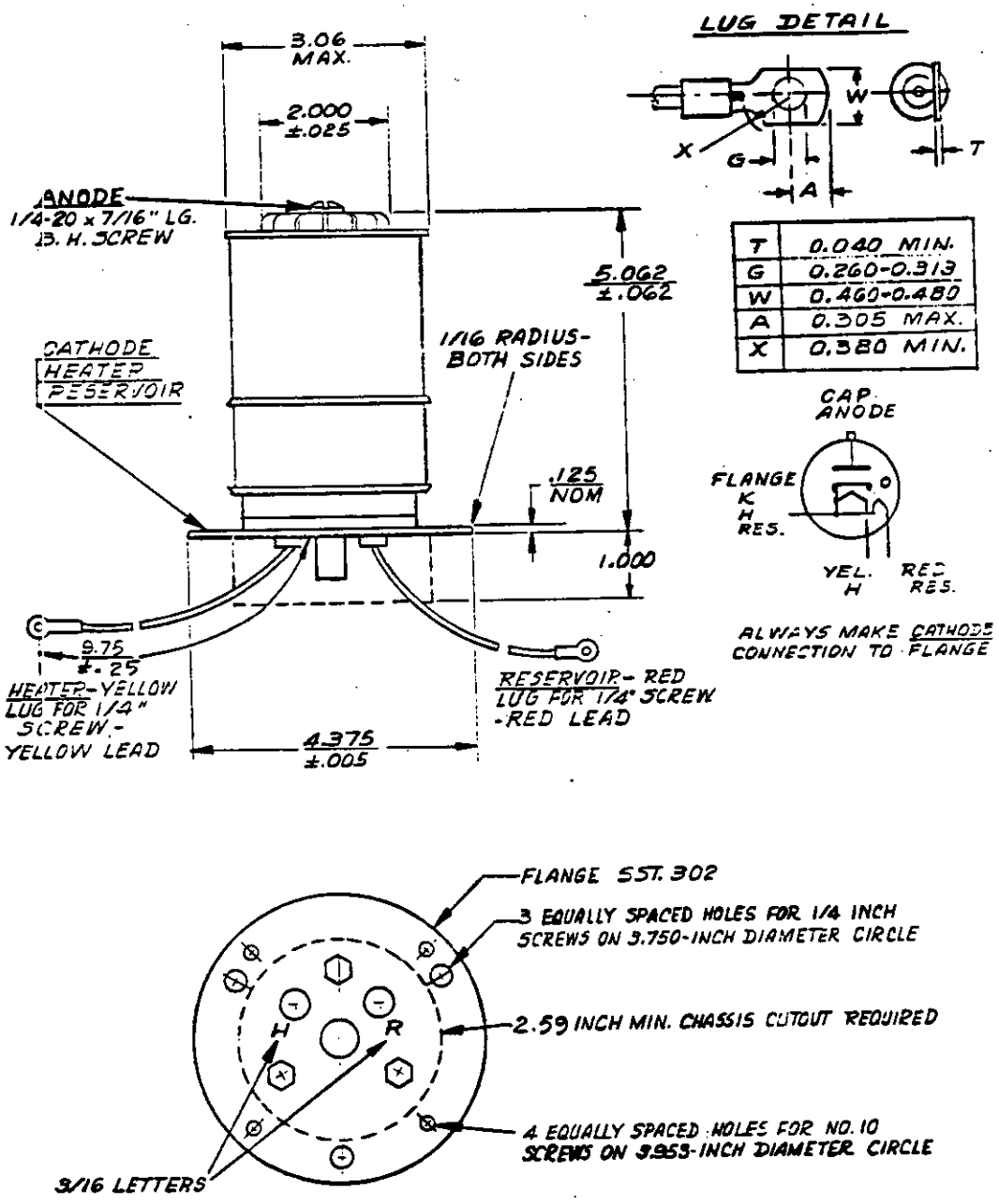
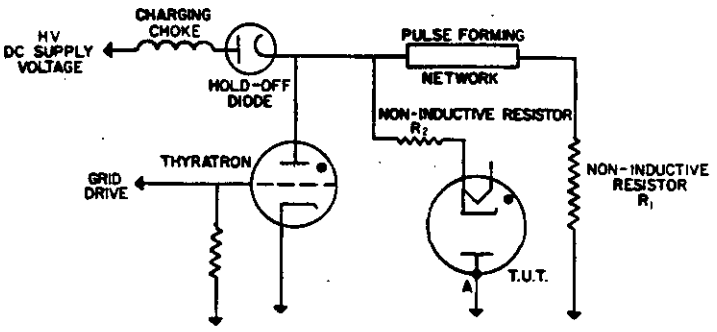


Figure 1 - Outline dimensions of tube type 8492.

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R_2 8.5 OHMS

Figure 2 - Test circuit.

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